

Title: LLM-Based Intelligent Personal Finance Advisor with Personalized Savings Prediction and Investment Recommendation

Abstract:

Managing personal finances has become increasingly challenging due to rising living expenses, changing spending habits, and the growing number of investment options available. Many individuals struggle to understand where their income is being spent, identify unnecessary expenses, and determine how much they can realistically save each month. Existing budgeting applications mainly track expenses and generate basic reports, but they often fail to provide personalized financial guidance or actionable recommendations tailored to an individual's spending behavior.

Furthermore, after identifying potential savings, users are often uncertain about how to utilize those savings effectively. Most financial platforms either focus solely on expense management or only provide investment-related information without considering the user's financial condition, income, spending patterns, and risk profile. As a result, users may miss opportunities to improve their financial health and make informed investment decisions.

To address these challenges, this project proposes an **LLM-Based Intelligent Personal Finance Advisor with Personalized Savings Prediction and Investment Recommendation**. The system leverages machine learning to analyze users' income and expenditure patterns, compare them with similar spending profiles, and predict achievable monthly savings. It further integrates live stock market data to recommend suitable investment opportunities based on the user's available savings. A Large Language Model (LLM) generates clear, personalized financial advice in natural language, enabling users to make better budgeting and investment decisions while improving their overall financial well-being.

Datasets:

Family Income and Expenditure Dataset:

Link: <https://www.kaggle.com/datasets/grosvenpaul/family-income-and-expenditure>

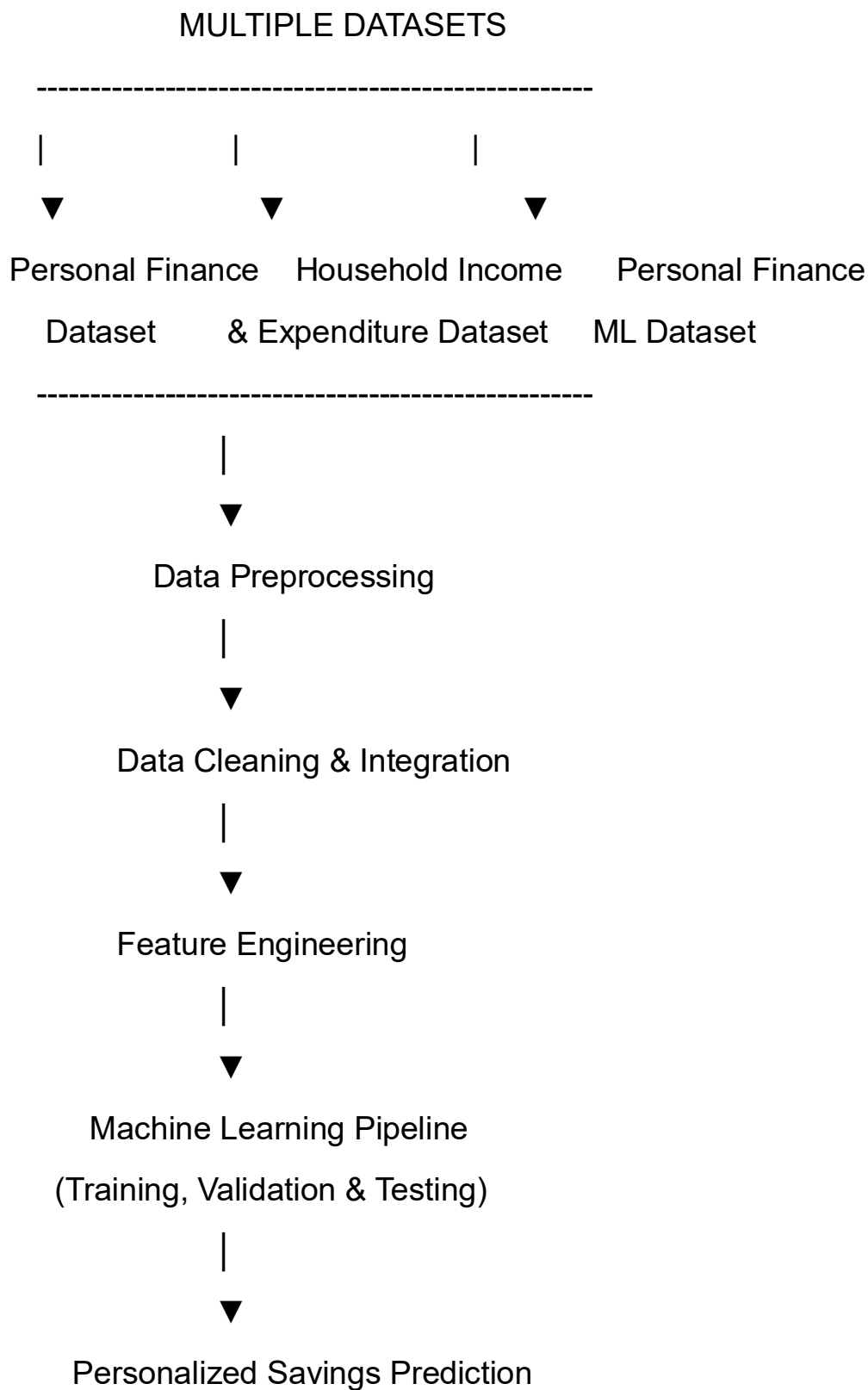
Personal Finance Data:

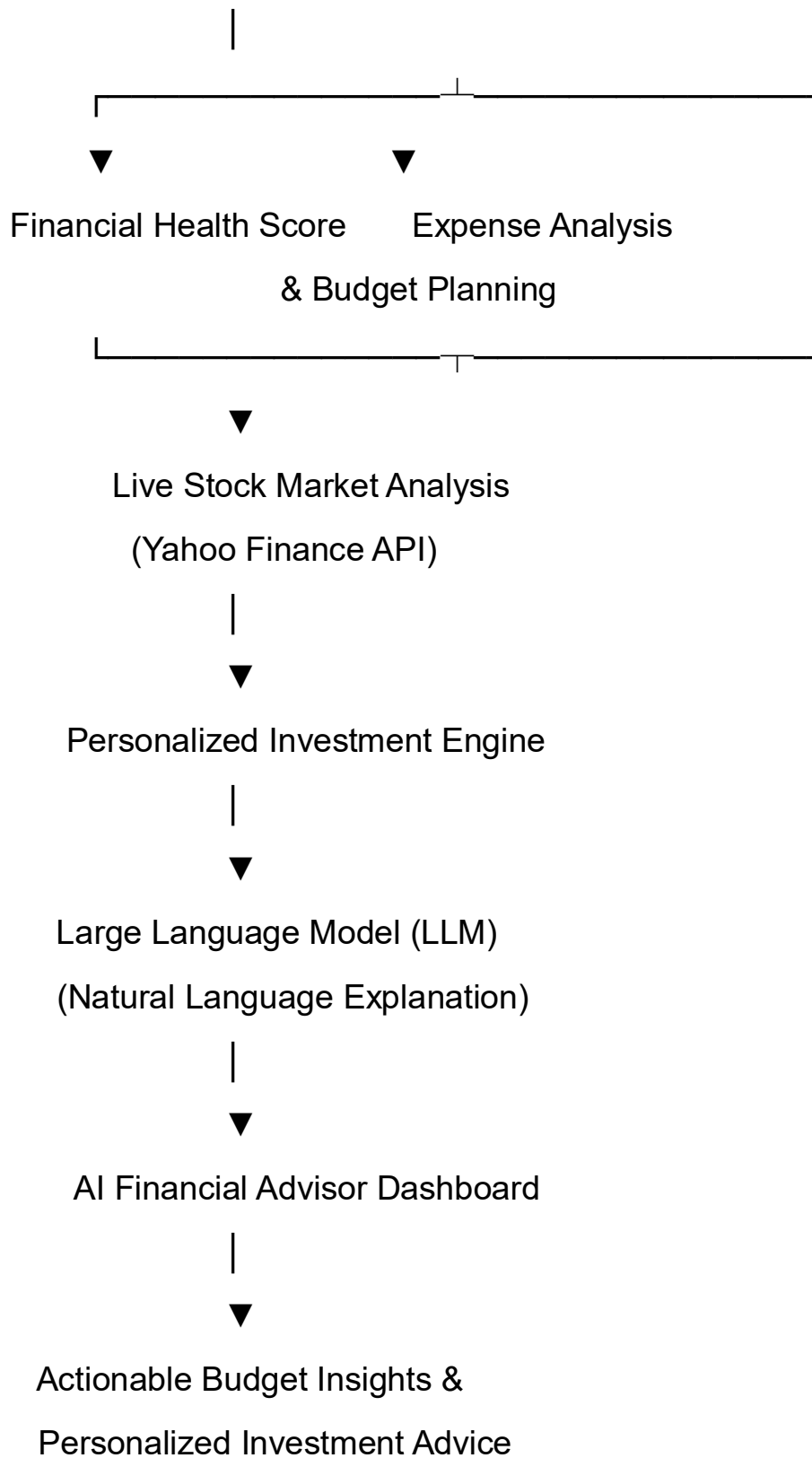
<https://www.kaggle.com/datasets/ramyapintchy/personal-finance-data>

Personal Finance ML Dataset:

<https://www.kaggle.com/datasets/miadul/personal-finance-ml-dataset>

Approach:





Project Group Members:

S.NO.	Roll Number	Student Name
1	2420030610	Ch. Srikar
2	2420030406	k. Narasimha Sai
3	2420080075	D. Hari Haran
4	2420080039	P. Jyothik Kshitij